

OPERATIONS RESEARCH

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS05	HSMC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
Contact Classes: 42	Tutorial Classes: 00	Practical Classes:--			Total Classes: 42			

COURSE OBJECTIVES:

To learn

1. Explain optimum utilization of resources, effort and implement the decision effectively.
2. Understand scientific systematic approach involved and provide a good intellectual support for decision making.
3. Discuss substantial experience in taking timely management decisions and for corrective measures.
4. Analyzing the behavior of the system for the purpose of improving its performance.
5. Demonstrate reliability of solution obtained from a model depends on the validity of the model in representing the real systems.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Applying a different linear programming problem technique which has a broad experience in finding the optimum profit.
2. Apply the knowledge of the course in solving real life problems.
3. Calculate the knowledge that tries to optimize total return by maximizing the profit and minimizing the cost or loss
4. Analyze the best (optimum) decisions relative to largest possible portion of the total organization.
5. Analyze queuing models and characteristic features of a queuing system.

UNIT-I	LINEAR PROGRAMMING	
Introduction to Linear Programming – Graphical method, Simplex methods, Primal & Dual Problem Sensitivity Analysis.		
UNIT-II	TRANSPORTATION AND ASSIGNMENT PROBLEMS	
Transportation problem – Methods of finding Initial Basic Feasible Solution, Test for optimality (MODI method), Assignment Problem –Hungarian Method ,Maximization case in assignment problem, unbalanced assignment problem.		
UNIT-III	GAME THEORY	
Decision theory-Decision under various conditions. Game Theory-Minimum & maximum strategies. Application of linear programming		
UNIT-IV	INVENTORY MODELS	
Stochastic inventory models-Single & multi period models with continuous & discrete demands. Cartesian co-ordinates).		
UNIT-V	QUEING MODELS	
Simulations-Monte Carlo simulation, generation of random numbers & simulation. Queuing models-M.M.1 & M/M/S system cost consideration		

TEXT BOOKS:

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| <ol style="list-style-type: none">1. H.A.Taha, — “Operations Research ”, Prentice Hall of India, 1999, Sixth Edition.2. J.K.Sharma, — “Operations Research Theory and applications”, Macmillian |
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REFERENCE BOOKS:

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| <ol style="list-style-type: none">1. R.Panner Selvam – “Operation Research”, PHI Learning Private Limited.2. Hillier and Lieberman, — “Operations Research ”, Holden Day, 1986 |
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